Subspace: Let V be a Vector space over the field F, and let $W \subseteq V$. Then W is called a subspace of V if W is also a vector space under the same operations of addition and scal multiplication as in V.

On the otherhand, A non-empty subset $W \subseteq V$ is a subspace it should satisfies the following condition.

1. Zero Vector $\bar{0} \in W$
2. Closure under Addition: $u \in W, v \in W$
   $u+v \in W$.
3. Closure under Scalar Multiplication:
   If $u \in W$ and $a \in F$
   then $au \in W$

Ex: The Set $W = \{(x,y) \in \mathbb{R}^2 \mid x+y=0\}$ is a Subspace.

If the Set $W = \{(x,y) \in \mathbb{R}^2 \mid x+y=0\}$ is a Subspace then it Shoud Satisfies above Conditions

1. The Set $W = \{(x,y) \in \mathbb{R}^2 \mid (x+y)=0\}$
   (i) Zero Vector $(0,0) \in W$ then $0+0=0 \in W$
   (ii) Closure under addition
   let $(x_1, y_1)\ (x_2, y_2) \in \mathbb{R}^2$
   then $(x_1+x_2,\ y_1+y_2) = (x_1+x_2+y_1+y_2) \in \mathbb{R}^2$
   $= 0$
   Since $(x_1, y_1) \in \mathbb{R}^2 \Rightarrow x_1+y_1=0$ $(x_2, y_2) \in \mathbb{R}^2 \Rightarrow x_2+y_2=0$.

(iii) Closed Under Scalar Multiplication:

$$a(x,y) = (ax, ay) = a(x+y)$$
$$= a(0)$$
$$= 0$$

Hence, the set $\{(x,y) \in \mathbb{R}^2 / x+y=0\}$ is a subspace

Prob: The set $W = \{(x,y) \in \mathbb{R} / x+y=1\}$ is it a subspace?

NO it is not a subspace

Zero vector $(0,0) \notin W$.

Prob: The set $W = \{f(x) = ax+b / a,b \in \mathbb{R}\}$ is a subspace?

If it is a subspace it should satisfies all three conditions of subspace

1. Zero Vector i.e., $f(0) = a \cdot 0 + b \in W$.

2. Closed Under addition.

$$f(x) = ax+b \qquad g(x) = cx+d.$$
$$(f+g)(x) = f(x)+g(x) = ax+b+cx+d$$
$$= (a+c)x + (b+d) \in W$$
Since $(a+c)(b+d) \in \mathbb{R}$.

3. closed Under Scalar Multiplication.

i.e, $a \in \mathbb{R}$ $f(x) = ax+b$.

$af(x) = a(ax+b) = a^2x + ab \in W$ $\because a,b,a^2 \in \mathbb{R}$.

$\therefore$ it is a subspace.

Prob: The set $W = \{(x_1, y_1, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 0\}$ is Subspace or Not.

No, it is not a subspace W is not a closed under addition i.e.,

let $x = (x_1\ x_2\ x_3) \in \mathbb{R}^3$ & $(y_1\ y_2\ y_3) \in \mathbb{R}^3$.

$\Rightarrow x_1^2 + x_2^2 + x_3^2 = 0$ $\quad y_1^2 + y_2^2 + y_3^2 = 0$

Now $x + y = (x_1, x_2, x_3) + (y_1, y_2, y_3)$

$= (x_1 + y_1, x_2 + y_2, x_3 + y_3)$.

$= (x_1 + y_1)^2 + (x_2 + y_2)^2 + (x_3 + y_3)^2 = 0$

By the given def of W.

$x + y = x_1^2 + y_1^2 + 2x_1y_1 + x_2^2 + y_2^2 + 2x_2y_2 + x_3^2 + y_3^2 + 2x_3y_3$.

$= x_1^2 + x_2^2 + x_3^2 + y_1^2 + y_2^2 + y_3^2 + 2x_1y_1 + 2x_2y_2 + 2x_3y_3$

$= 0 + 0 + 2(x_1y_1 + x_2y_2 + x_3y_3) \neq 0$.

$\Rightarrow (x + y) \in W$.

$\therefore$ It is not a subspace.

Prob: The given set $W = \{(x,y,z) \in \mathbb{R}^3 / x-2y+3z=0\}$ is a subspace?

The given set $W = \{x,y,z \in \mathbb{R}^3 / x-2y+3z=0\}$ is a subspace it should satisfies all three conditions of subspace.

i.e.,

1. zero vector $(0,0,0) \in \mathbb{R}^3 \Rightarrow 0-2\cdot 0+3\cdot 0 = 0 \in W$.

2. Closed under addition

Let $x = (x_1, x_2, x_3) \in \mathbb{R}^3$

$y = (y_1, y_2, y_3) \in \mathbb{R}^3$ By def.

$x_1 - 2x_2 + 3x_3 = 0$

Ily $y_1 - 2y_2 + 3y_3 = 0$

Now $x+y = (x_1 - 2x_2 + 3x_3) + (y_1 - 2y_2 + 3y_3)$

$= (x_1+y_1) - 2(x_2+y_2) + 3(x_3+y_3)$.

$= (x_1 - 2x_2 + 3x_3) + (y_1 - 2y_2 + 3y_3)$

$= 0+0 = 0$

$x+y \in W$.

3. W closed under Scalar Multiplication

$ax = a(x_1, x_2, x_3) = (ax_1, ax_2, ax_3)$ from (i)

$= ax_1 - 2ax_2 + 3ax_3$.

$= a(x_1 - 2x_2 + 3x_3)$.

$= a \cdot 0 = 0$

$ax \in W$.

Prob: Set of all continuous functions $f: \mathbb{R} \to \mathbb{R}$ is a subspace.

Set of all continuous functions defined $f: \mathbb{R} \to \mathbb{R}$ is a subspace because addition and scalar multiplication of continuous function also continuous function

Prob: The set $W = \{ f \in [0,1] \mid f(0) = 1 \}$ is not a vector space or subspace.

It is not a vector space or a subspace because W is closed under addition

i.e., let $f, g \in [0,1] \Rightarrow f(0) \geq 1$, $g(0) \geq 1$.

$(f+g)(0) = f(0) + g(0)$

$= 1+1 = 2 \neq 1 \notin W$.

$(f+g) \notin W$

Prob: The set $W = \{ A \in M_{2\times 2}(\mathbb{R}) \mid \text{trace } A = 0 \}$ is a subspace.

$W = \{ A \in M_{2\times 2}(\mathbb{R}) \mid \text{trace } A = 0 \}$ is a subspace it satisfies all three conditions.

i) $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in M_{2\times 2}(\mathbb{R})$ then $tr(A) = 0+0 = 0$.

$A \in W$ i.e., Zero Matrix in W.

(ii) Let $A, B \in M_{2\times 2}(\mathbb{R})$ then $tr(A) = 0$ $tr(B) = 0$ by defn of W

Now $tr(A+B) = tr\,A + tr\,B$

$tr(A+B) = 0+0 = 0 \in W$.

(iii) W is closed under scalar Multiplication.

i.e., $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2}(\mathbb{R})$ then $tr\,A = a+d = 0$.

Now $\alpha \in \mathbb{R}$ $\alpha A = \alpha \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{bmatrix}$

$\alpha A = tr(\alpha A) = \alpha a + \alpha d = \alpha(a+d) = \alpha \cdot 0 = 0$.

$\alpha A \in W$.

$\therefore$ W is a subspace.

Prob: The set of all even functions $W = \{f : R \to R \mid f(x) = f(x)\}$ is a subspace. $f(-x) = f(x)$

W is a satisfies all three conditions of subspace

i.e (i) Zero function is belong to W.

(ii) Let $f, g \in W$ $f(-x) = f(x)$ $g(-x) = g(x)$

Now $(f+g)(-x) = f(-x) + g(-x)$

$= f(x) + g(x)$.

$\Rightarrow (f+g) \in W$.

W is closed under addition

(iii) W is closed under scalar Multiplication

i.e., $\alpha \in \mathbb{R}$ $f \in W$

$(\alpha f)(-x) = \alpha f(-x) = \alpha f(x)$

$= (\alpha f)(x)$.

$\Rightarrow \alpha f \in W$.

Similarly the set of all odd functions $W = \{f : R \to R / f(-x) = f(x)\}$ $f(-x) = -f(x)$

is also a subspace.

Prob: The Set $N = \{(x,y) \in \mathbb{R}^2 / xy = 0\}$ is not a Vector space and subspace.

The Set $N = \{(x,y) \in \mathbb{R}^2 / xy = 0\}$ is not a Vector space and subspace, because W is not closed under addition i.e.,

let $(x,y) \in \mathbb{R}^2$

$(x_1,x_2) \in \mathbb{R}^2 \Rightarrow x_1x_2 = 0$ by def^n of W

$(y_1,y_2) \in \mathbb{R}^2 \Rightarrow y_1y_2 = 0$. by def^n of W

then $(x+y) = (x_1,x_2) + (y_1,y_2) = [(x_1+y_1), (x_2+y_2)$

$= x_1x_2 + x_1y_2 + x_2y_1 + y_1y_2$

$= x_1y_2 + y_1x_2 \neq 0$.